# Project Initialization and Planning Phase

| | |
|---|---|
| Date | 18 July 2026 |
| Team ID | TBD |
| Project Title | Voice Based Concept Understanding Analyser |
| Maximum Marks | 5 Marks |

### Project Proposal (Proposed Solution) report

The proposed solution is an AI-powered Voice Based Concept Understanding Analyser that evaluates a student's conceptual understanding through spoken responses. The system converts speech into text using Whisper, compares the response with the expected answer using Sentence Transformers, generates AI-based feedback using Gemini, stores the results in SQLite, and creates downloadable PDF reports for students and educators.

| **Project Overview** | |
|---|---|
| Objective | To automatically evaluate students' conceptual understanding from voice responses and provide accurate scores, semantic similarity analysis, personalized AI feedback, and downloadable performance reports. |
| Scope | The project supports educational institutions, teachers, and students by automating concept evaluation using Artificial Intelligence, Natural Language Processing, Speech Recognition, and Semantic Similarity techniques. |
| **Problem Statement** | |
| Description | Traditional oral examinations and manual assessments are time-consuming, subjective, and inconsistent. Teachers often struggle to evaluate students individually and provide immediate feedback. There is a need for an AI-based solution that can automatically analyze spoken answers and generate accurate evaluations. |

| | |
|---|---|
| Impact | The proposed solution reduces manual effort, improves evaluation accuracy, provides instant personalized feedback, enhances learning outcomes, and saves valuable time for both students and educators. |
| **Proposed Solution** | |
| Approach | The system records the student's voice response, converts it into text using Whisper, compares it with the expected answer using Sentence Transformers, generates AI feedback through Gemini, stores all results in SQLite, and produces a PDF performance report using ReportLab. |
| Key Features | • Voice-to-text conversion using Whisper<br>• Semantic similarity evaluation<br>• AI-generated feedback using Gemini<br>• Automatic scoring system<br>• Performance dashboard<br>• SQLite database storage<br>• PDF report generation<br>• User-friendly Streamlit interface |

SMARTBRIDGE Let's Bridge the Gap

SKILL WALLET A SmartBridge Product

**Resource Requirements**

| **Resource Type** | **Description** | **Specification/Allocation** |
|---|---|---|
| **Hardware** | | |
| Computing Resources | CPU/GPU specifications, number of cores | Intel Core i5 or above |
| Memory | RAM specifications | 8 GB RAM |

| | | |
|---|---|---|
| Storage | Disk space for data, models, and logs | 512 GB SSD |
| **Software** | | |
| Frameworks | Python frameworks | Streamlit |
| Libraries | Additional libraries | Whisper, Sentence Transformers, SQLAlchemy, ReportLab, Pandas, NumPy |
| Development Environment | IDE | VS Code |
| **Data** | | |
| Data | Source, size, format | Student voice recordings and reference answers |